

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st November 2024

SA2 - Life Insurance

Time allowed: 3 Hours 15 Minutes (14.45 – 18.00 Hours)

Total Marks: 100

- Q. 1)** A medium sized Indian life insurance company sells products across multiple lines of business including participating, non-participating and unit-linked business.

It has recently appointed a new Appointed Actuary.

- a)** In addition to the general duties and obligations of Appointed Actuary of all insurers, what are the additional duties and obligations of an Appointed Actuary specifically for an insurer carrying on life insurance business as prescribed by the IRDAI regulations in respect of Appointed Actuaries? (7)

There is a proposal to invest up to 15% in equities under individual non-participating savings business.

- b)** Prepare a detailed note setting out the relative merits/demerits of adopting the proposal. In your note, you should, as a minimum, cover the following:

- I.** A description of the salient features of individual non-participating savings product as generally available in India; (2)

II. Implications of the proposal on each of the following:

- (i)** Adherence to regulatory limits and constraints; (2)

- (ii)** Asset Liability Matching; (4)

- (iii)** Reserving, Solvency, Profit and Value assessments (EV/VNB). (8)

The company has also written unit linked contracts where the only charges were fund management charge, mortality charge and exit charges on early surrender. The death benefit is sum assured (or fund value if higher) as chosen by policyholder and maturity benefit is fund value. In addition, the plan provides loyalty addition of return of mortality charges only at maturity. The product is compliant with unit linked regulations applicable for Indian market and it is open to new business.

- c)** Describe the main risks faced by the company under this product. (16)

- d)** Establish a process of reserving for loyalty additions. (11)

[50 Marks]

- Q. 2)** An Indian life insurance company sells all lines of business but primarily sells participating business - both life and pensions; and values the business using Indian Embedded Value (IEV).

- a)** List the minimum information to be provided while disclosing the analysis of change in IEV as per the Actuarial Practice Standard 10 (APS 10) of the Institute of Actuaries of India. (8)

- b)** Briefly state the primary constraint within the participating product construct that might result in margins for individual conventional participating savings endowment plans being potentially lower than that of non-participating plans and state what actions can be taken to increase the profits under participating contracts including their implications. (10)

The company has a separate individual with-profits pensions fund. The company has historically written a significant amount of deferred pension business in this fund but have stopped writing new business for past 10 years. The fund is in run-off mode. Currently, the fund has accumulated a significant amount within the Funds for Future Appropriation (FFA).

Given run-off business and that the historical FFA accumulated is significant, the actuary is exploring ways to manage this FFA.

One of the options suggested was to distribute the FFA.

- c) List the implications (challenges, risks and opportunities) of distribution of FFA. (13)

Another option suggested is to assess feasibility of merging the with-profits pension fund with the Company's with-profit life fund.

- d) List the additional challenges of merging the pension with-profits fund with the life with-profits fund. (7)

The company is exploring reinsurance arrangements to manage its mortality risk.

- e) Explain briefly the general considerations to be looked at in a reinsurance arrangement by a life insurance company. (12)

[50 Marks]
